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Critical Issues of Fluorinated Alkoxyborate–Based Electrolytes in Magnesium Battery Applications

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Abstract

The development of non-corrosive but highly efficient electrolytes has been a long-standing challenge in magnesium rechargeable battery (MRB) research fields. As fluorinated alkoxyborate-based electrolytes have overcome serious problems associated with conventional electrolytes, they are regarded as promising for practical MRB applications. An electrolyte containing representative magnesium fluorinated alkoxyborate $Mg[B(HFIP)_4]_2$ ($[B(HFIP)_4]$: tetrakis(hexafluoroisopropoxy) borate) was prepared through general synthetic routes using $Mg(BH_4)_2$, however, it shows poor electrochemical magnesium deposition/dissolution behavior. Herein, we report an alternative synthetic route of highly reactive $Mg[B(HFIP)_4]_2$ and several critical issues associated with the use of $Mg[B(HFIP)_4]_2$/glyme electrolytes in MRBs. The cycling performance of the electrolytes as well as the synthetic reproducibility of the salt was significantly improved upon adopting a transmetalation reaction between certain magnesium and boron compounds for the salt preparation. Despite the outstanding electrochemical activities of $Mg[B(HFIP)_4]_2$/glyme, the electrolytes were unstable with magnesium metal. The remarkably high dissociativity of $Mg[B(HFIP)_4]_2$ in glyme solutions and resulting enhanced induction interaction of Mg^{2+} with coordinated glymes makes the solutions reductively unstable. Surface passivation by [TFSA]-based electrolytes (TFSA: bis(trifluoromethanesulfonyl)amide) effectively suppressed the decomposition of $Mg[B(HFIP)_4]_2$/glyme electrolytes. This passivation simultaneously caused large overpotential for electrochemical cycling. The short-circuiting of the cells upon repeated deposition/dissolution cycling is rather problematic. The findings disclose here the issues of fluorinated alkoxyborate-based electrolyte solutions that should be resolved for practical MRB materialization. We also emphasize

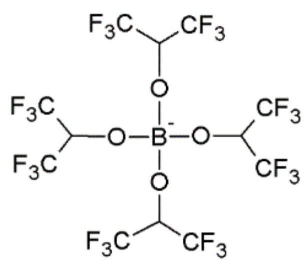
the importance of systematic strategies in manipulating the electrolytes and interfaces as well as base magnesium metal based on each appropriate approach.

Introduction

Multivalent rechargeable batteries—specifically those based on earth-abundant, reductive metals such as Mg(II), Ca(II), and Al(III)—are candidates for next-generation energy storage technologies owing to the potentially high energy storage performance and cost-effectiveness of large-scale battery production.[1,2] The energy densities of the prototypes of these multivalent batteries are, however, far less than those of current lithium-ion batteries. The initial prototype of magnesium rechargeable batteries (MRBs) proposed in 2000 delivered ca. 90 mAh g^{-1} at 1.1 V of working voltage, corresponding to ca. 100 Wh kg^{-1} of energy density based only on the mass of the cathode active material Mo_6S_8.[3] Since this pioneering work, research into novel materials for high energy density MRBs has progressed. Significant progress has been achieved in both cathode active materials and electrolytes in the last 5 years.[4–14] However, the research stages of MRBs are still far from practical materialization, mainly due to the lack of suitable electrolytes compatible with magnesium anode, practical cathode active materials, and other battery components.

The development of non-corrosive but highly efficient electrolytes has been long-standing challenges in MRB research fields. In recent years, a new class of electrolyte materials especially those incorporating weakly coordinated anions (WCAs) have been developed.[8–14] The electrochemical properties of these electrolytes are comparable to those of organic haloaluminate-based electrolytes despite the absence of any reactive activation reagents, which has caused safety issues under battery operations. Certain fluorinated alkoxyborate or alkoxyaluminate-based electrolytes are regarded as promising candidates for practical MRB materialization among reported WCA-based electrolytes.[9,10,11,13,14] Owing to their non-nucleophilic nature combined with favorable electrochemical magnesium redox activity and oxidative stabilities, they have attracted considerable attention as a primary electrolyte option especially for magnesium sulfur batteries.[10,14–16]

WCA-based electrolytes were first reported several years ago. The limited number of studies on WCA-based electrolytes that have been reported so far is likely due to the nontrivial synthesis of WCA salts applicable to electrolytes. Magnesium salts incorporating carborane structures such as $Mg(CB_{11}H_{12})_2$ required expensive reagents and multi-step reactions to synthesize, however, this salt imparts remarkably favorable electrochemical properties.[8] Synthetic procedures of fluorinated alkoxyaluminate salts have been investigated. Their syntheses either involve several steps of reactions or require special experimental set-ups and commercially unavailable starting materials.[9,13,17] In contrast to these aforementioned WCA salts, fluorinated alkoxyborate can be obtained easily by the single-step dehydrogenation of commercially available $Mg(BH_4)_2$ by certain fluorinated alcohols.[10] The synthesis of representative magnesium fluorinated alkoxyborate salt, $Mg[B(HFIP)_4]_2$ ($B(HFIP)_4$: tetrakis(hexafluoroisopropoxy)borate, Scheme 1), from commercial $Mg(BH_4)_2$ sometimes does occur satisfactorily. The electrochemical performance for the resulting salts depends on the reaction batch. Such situations make deeper studies on this type of salts difficult.



Scheme 1. Chemical structure of $[B(HFIP)_4]^-$ anion.

Herein, we report an alternative synthetic route of highly reactive $Mg[B(HFIP)_4]_2$. The cycling performance of electrolytes, as well as the synthetic reproducibility of the salt, was significantly improved upon adopting a transmetalation reaction between certain magnesium and boron compounds, instead of the simple dehydrogenation of $Mg(BH_4)_2$ for the salt preparation. Several critical issues associated with the use of $Mg[B(HFIP)_4]_2$/glyme electrolytes in MRBs are also discussed and possible strategies are proposed.

Experimental

Materials

Magnesium borohydride ($Mg(BH_4)_2$; 95%) and di-*n*-butyl magnesium ($Mg(Bu)_2$; 1.0 mol dm^{-3} heptane solution) were purchased from Sigma-Aldrich and used as received. Trimethylalmunium (15% in toluene) and a respective Grignard reagent ca. 2 mol dm^{-3} C_2H_5MgCl/THF were purchased from Tokyo Chemical Industries, Co., Ltd. 1,1,1,3,3,3-Hexafluoroisopropanol (HFIP-H), tetrahydrofuran-borane tetrahydrofuran solution (BH_3-THF), and deuterated dimethyl sulfoxide (DMSO-*d*6) were purchased from FUJIFILM Wako Chemicals. HFIP-H was distilled over calcium oxide and stored in an Ar-filled glovebox with 3A molecular sieves. Monoglyme (G1), diglyme (G2), and triglyme (G3), for electrochemistry, were obtained from Kanto Chemical Co., Inc., and used without further purification. Battery-grade magnesium bis(trifluoromethanesulfonyl)amide ($Mg[TFSA]_2$) was purchased from Kishida Chemical Co., Ltd. $Mg[TFSA]_2$ was dried under vacuum heating at 120 °C for several days and stored in an Ar-filled glovebox prior to use. The electrolyte solutions were prepared by mixing the prefixed amounts of $Mg[B(HFIP)_4]_2$ or $Mg[TFSA]_2$ in glymes, followed by vigorous stirring at 30 °C overnight in the glovebox (< 1 ppm H_2O and O_2). For the cloudy solution containing $Mg[B(HFIP)_4]_2$ made from $Mg(BH_4)_2$, the insoluble materials were filtered to obtain a clear solution. The water content of the prepared electrolytes measured by Karl Fischer titration was confirmed as less than 50 ppm.

Syntheses

$Mg[B(HFIP)_4]_2$ was synthesized according to two different procedures in this study. As complete removal of solvated G1 from $Mg[B(HFIP)_4]_2{\cdot}3G1$ was extremely difficult, we used the adduct as a supporting salt of the electrolyte solutions, and hereafter denoted it simply as $Mg[B(HFIP)_4]_2$.

Procedure 1, from $Mg(BH_4)_2$: HFIP-H (10.0 eq., 19.2 mL, 185 mmol) was added dropwise to the solution of $Mg(BH_4)_2$ (1.0 g, 18.5 mmol) dissolved in anhydrous G1 (100 mL) over a period of 1 h at 0 °C. H_2 gas was generated immediately. After H_2 generation completed, the reaction mixture was stirred at 25 °C for 24 h. Then, the solvent and excess HFIP-H were removed by vacuum heating at 45–50 °C for 24 h yielding 6.9–27.8 g (22.6–91.0%) of the product as a white powder. The ^1H nuclear magnetic resonance (NMR) spectrum of the product suggests the formation of $Mg[B(HFIP)_4]_2{\cdot}3G1$ adducts (Figure S1). ^1H NMR (DMSO-*d*6, 400 MHz): δ (ppm) = 3.223 (s, $CH_3OCH_2CH_2OCH_3$, 18H), 3.413 (s, $CH_3OCH_2CH_2OCH_3$, 12H), 4.632 (m, $(CF_3)_2CHO$, 8H).

Procedure 2, from $Mg(Bu)_2$ and BH_3-THF: HFIP-H (2.05 eq. *vs*. Mg, 2.13 mL, 20.5 mmol) was added to the solution of $Mg(Bu)_2$/heptane (10.0 mL, 10.0 mmol) dropwise over a period of 10 min at 0 °C. The solid product, $Mg(HFIP)_2$, was dissolved in 20 mL of G1 and the resulting solution was stirred at 25 °C for 30 min. BH_3-THF (2.2 eq. *vs*. Mg, 24.4 mL, 22 mmol) was added dropwise into the $Mg(HFIP)_2$/G1 solution over a period of 30 min at 25 °C, followed by the further addition of HFIP-H (3.1 eq. vs. B, 7.07 mL, 68.2 mmol) dropwise over 1 h at 25 °C. The mixture was vigorously stirred for 24 h at 25 °C. After that, the solvent and excess HFIP-H were removed by vacuum heating at 50 °C for 24 h, yielding 14.9–15.6 g (90.2–94.4%) of the product as a microcrystalline powder. The ^{1}H NMR spectrum of the product suggests the formation of $Mg[B(HFIP)_4]_2·3G1$ adducts (Figure S1). For the $Mg[B(HFIP)_4]_2$ syntheses, various active Mg and B sources, such as $Mg(CH_3)_2$, $Mg(C_2H_5)_2$, BH_3-$S(CH_3)_2$, BH_3-morpholine complexes, were also applicable. ^{1}H NMR (DMSO-*d*6, 400 MHz): δ (ppm) = 3.223 (s, $CH_3OCH_2CH_2OCH_3$, 18H), 3.414 (s, $CH_3OCH_2CH_2OCH_3$, 12H), 4.623 (m, $(CF_3)_2CHO$, 8H).

Measurements

NMR spectra were recorded in DMSO-*d*6 using a JEOL JNM-ECA 400 spectrometer with ^{1}H resonance frequency of 400 MHz. The chemical shifts of ^{1}H signals were obtained relative to the residual DMSO peak as the internal standard.[18] Typical magnesium deposition-dissolution cycling tests were conducted by cyclic voltammetry (CV) with a three-electrode beaker cell. A Pt disk (φ3 mm, BAS), a magnesium ribbon (FUJIFILM Wako Chemicals) were used as working and counter electrodes, respectively. A Ag^+/Ag reference electrode was fabricated according to a procedure described elsewhere.[19] The electrode potential of the reference electrode in a series of $Mg[B(HFIP)_4]_2$/glyme was calibrated using 0.3 mol dm^{-3} $Mg[Al(HFIP)_4]_2$/G2 to be −2.49 V *vs*. the reference, which corresponds to 0 V *vs*. Mg^{2+}/Mg^0. The 0.3 mol dm^{-3} $Mg[Al(HFIP)_4]_2$/G2 solution was prepared by mixing prefixed amounts of $Mg(HFIP)_2$ and $Al(HFIP)_3$ in G2 followed by continuous stirring for 3 days at 25 °C. The CV curve of the Mg electrode (counter: Pt coil, reference: Ag^+/Ag) recorded in this electrolyte is shown in Figure S2. All electrochemical cells were assembled in the glovebox and the CV measurements were conducted using an electrochemical analyzer HSV-110 (Hokuto-Denko) located in the glovebox. A galvanostatic magnesium deposition/dissolution cycling test was performed on the two electrode cells where Cu (Nilaco) and AZ31 alloy (Nippon Kinzoku CO., Ltd.) foil were employed as a working and counter electrode, respectively. Three different separators, GA200 (ADVANTEC, t = 0.74 mm), GF/A (Whatmann, t = 0.26 mm) and three-dimensionally-ordered microporous polyimide[20] (3DOM; t = 0.03 mm) were used in this study. A two-electrode-type cell was assembled in the glovebox. A galvanostatic cycling test was carried out at a current density of 0.5 mA cm^{-2} for 30 min for each deposition and dissolution at 30 °C using an automatic charge/discharge instrument (HJ0610SD8Y, Hokuto Denko). All electrochemical measurements were conducted at least three times and the results were confirmed to be reproducible.

The ionic conductivities of the electrolytes were determined by the complex impedance method using an impedance analyzer (VMP3, Biologic) in the frequency range of 500 kHz–1 Hz with a sinusoidal alternating voltage amplitude of 10 mV root-mean-square (rms). A cell equipped with two platinized platinum electrodes (CG–511B, TOA Electronics), whose cell constant was determined using a 0.01 M KCl aqueous solution at 25 °C, was used for the impedance measurements. The cell was placed in a temperature-controlled chamber and held at the prefixed temperature for 1 h to reach the temperature equilibrium; then, the conductivity was measured at each desired temperature. Raman spectra of the magnesium salts and electrolyte solutions were collected with a Nicolet iS50R FT-Raman spectrometer (Themo Scientific) equipped with a 1064-nm laser at a resolution of 1 cm^{-1}. The spectrometer was placed in a dry room (dew point −60 ºC) to avoid exposing the samples to the ambient air atmosphere. The spectra were all calibrated by the polypropylene standard and Si single crystal. The surfaces and cross-section of the magnesium metal and separator were observed with scanning electron microscopy (SEM; JSM-7800F, JEOL) and subsequently characterized by energy dispersive X-ray spectroscopy (EDX), X-ray photoelectron spectroscopy (XPS; VersaProbe II, ULVAC-PHI), and auger electron spectroscopy (AES; PHI710, ULVAC-PHI). The separator was processed by a focused ion beam to obtain a smooth cross-sectional view. The samples were all washed with anhydrous THF to remove the residual electrolyte and subsequently dried under high-vacuum conditions, then placed in an airtight chamber, and transferred for various analysis without any exposure to air. XPS measurements were performed with an Al Kα X-ray source under a base pressure of 6.7×10^{-8} Pa. Depth profiles were taken by argon sputtering with an ion gun. Emission currents of 20 mA and an acceleration voltage of 2 kV were applied. The etching time was 0–100 min in every 10 min. For spectroscopic analysis, the spectra were deconvoluted by Voigt function using appropriate Gaussian-Lorentzian ratios. The typical field emission phenomenon was utilized to excite the samples under a base pressure of 6.7×10^{-8} Pa for AES measurements.

Density functional theory (DFT) cluster calculation was performed to estimate the HOMO-LUMO energy levels of the free isolated $[B(HFIP)_4]^-$ and the anion coordinated to Mg^{2+} ($Mg[B(HFIP)_4]_2$) in G3 solvent. DFT molecular dynamics sampling by Car-Parrinello molecular dynamics (CPMD) simulations were carried out to obtain the relative equilibrium structures coordinated by large bulky molecules, using PBE-D2 functional and Goedecker-Teter-Hutter (GTH) pseudopotentials.[21] Then, Gaussian16 calculations with integral equation formalism polarizable continuum model (IEFPCM) for G3 (ε = 7.62) were applied to obtain the electronic states of [B(HFIP)4] and Mg[B(HFIP)4]2 taking G3 solvation effects into account. These calculations were performed by Gaussian16 Revision A.03 code,[22] using M06 hybrid functional and 6-311++G(d,p) basis sets.[23]

Results and Discussions

Synthesis and Basic Electrochemistry of $Mg[B(HFIP)_4]_2$: Many researchers have devoted their efforts to achieve highly efficient but non-corrosive MRB electrolytes for several decades. Halide, representatively chloride, containing electrolytes in general exhibit favorable electrochemical activity, while causing severe corrosion of various battery substrates and undesired dissolution of cathode active materials. The development of fluorinated alkoxyborate-based magnesium salts, e.g. $Mg[B(HFIP)_4]_2$, is significant progress in the field of magnesium rechargeable battery because of their excellent electrochemical activity, less corrosive nature, and acceptable oxidative stability of their oligo-ether solutions.[10] Despite such outstanding electrochemical characteristics, the synthesis of $Mg[B(HFIP)_4]_2$ and its application in MRB involves serious issues. We have attempted to synthesize $Mg[B(HFIP)_4]_2$ from $Mg(BH_4)_2$ several times; however, the reproducibility of the synthetic yields has been very poor, ranging from *ca*. 20 to 90%, even though the same synthetic procedure was adopted. In addition, the reaction batch also has a significant impact on electrochemical magnesium deposition/dissolution activities. Figure 1 displays the typical CV curves of the Pt electrodes recorded in 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3. For the CV curves of $Mg[B(HFIP)_4]_2$ from $Mg(BH_4)_2$, the profiles and current densities of magnesium deposition/dissolution strongly depend on the reaction batch, as an approximately 10 times larger current density of magnesium deposition at −1.0 V vs. Mg^{2+}/Mg was observed for batch 2 compared to that of batch 1. Such a severe synthetic situation impedes further researches using these types of electrolytes.

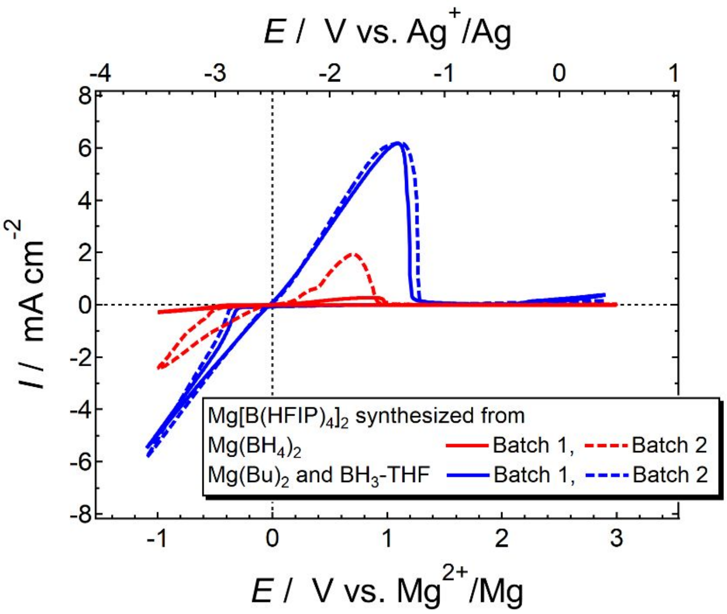


Figure 1. Typical CV curves of 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3 recorded on Pt electrode at 50 mV s^{-1} and 30 °C. Mg salts used were synthesized from the different starting materials.

To overcome this situation, different approaches should be developed. As the $Mg(BH_4)_2$ produced by solid-state and chemical syntheses can involve unspecified impurities such as MgH_2, MgX_2 (X = Cl or Br), and parent boron compounds,[24] a reaction type of transmetalation has been adopted to synthesize Mg[B(HFIP)$_4$]$_2$ (Scheme 2). Transmetalation is a well-known elementary reaction that involves the transfer of ligands between two metals and is often adopted for preparing typical MRB electrolytes, such as organohaloaluminate-based electrolytes where the transfer of an alkyl group from Mg to Al takes place.[3] Although isolation of B(HFIP)$_3$ from BH_3-THF and HFIP-H mixtures was failed due to unexpected supramolecular formation among B(HFIP)$_3$ and HFIP-H (data not shown), the transfer between Mg and B successfully proceeds to be desired magnesium salt by combining magnesium fluorinated alkoxide and in situ generated fluorinated alkoxyborate as a one-pot synthetic process. A similar approach applies to the Ca salt preparation.[25] The transmetalation worked particularly well for synthesizing $Mg[B(HFIP)_4]_2$ from appropriate Mg and B sources. As shown in Figure S1, the products obtained by different synthetic procedures provided the same ^1H NMR spectra, implying successful preparation of $Mg[B(HFIP)_4]_2$ via transmetalation and complete removal of any possible intermediates by vacuum heating owing to highly volatile or sublimable nature of starting reagents and possible intermediates. The comparison among the CV curves of the electrolyte solutions where BH_3-THF, HFIP-H, and $B(HFIP)_3$ were added intentionally further supports absence of these impurities in the products of transmetalation (Figure S3). Most importantly, the reactivity and reproducibility of the product significantly improved by using this procedure. As shown in Figure 1, the current densities and overpotential of electrochemical magnesium deposition/dissolution were improved compared to the electrolytes using the $Mg[B(HFIP)_4]_2$ salt from $Mg(BH_4)_2$. The two CV curves of the electrolytes incorporating $Mg[B(HFIP)_4]_2$ synthesized upon transmetalation showed almost identical profiles even though the salts from different batch were used, indicating the high reproducibility of the salt synthesis. The excellent electrochemical magnesium deposition/dissolution behavior on the Pt electrode was also observed for the electrolyte solutions of $Mg[B(HFIP)_4]_2$/glyme irrespective of the glyme length (Figure S4). Furthermore, the electrolyte did not require the pre-cycling conditioning process for activation (Figure S5). This feature seems favorable for the use of the $Mg[B(HFIP)_4]_2$ salt obtained by transmetalation in practical MRB applications.

$$Mg(Bu)_2 + 2HFIP\text{-}H \rightarrow Mg[HFIP]_2 + 2Bu\text{-}H$$

$$Mg[HFIP]_2 + 2BH_3\text{-}THF + 6HFIP\text{-}H \rightarrow Mg[B(HFIP)_4]_2 + 6H_2$$

Scheme 2. Proposed synthetic route to obtain $Mg[B(HFIP)_4]_2$ via transmetalation between Mg and B.

Although the electrochemical activities of the $Mg[B(HFIP)_4]_2$/glyme electrolytes strongly depend on whether the salt was synthesized by simple dehydrogenation of $Mg(BH_4)_2$ or transmetalation between corresponding Mg and B compounds, the morphology and atomic abundance of the deposits obtained on Pt electrode by potentiostatic polarization at −0.5 V vs. Mg^{2+}/Mg for 1 h were comparable irrespective of the salt sources (Figure S6). The difference in the size of second particles may arise from the difference in the current (charge) passed as the current density of the electrolytes containing the salt synthesized via transmetalation is 3 times larger than that containing the salt from $Mg(BH_4)_2$ at the same electrode potential (Figure 1). A minor, evident difference was found in their Raman spectra. A small peak was discernible at 748 cm^{-1} for the $Mg[B(HFIP)_4]_2$ salt from $Mg(BH_4)_2$ while no peak was found at this wavenumber for the salt synthesized by transmetalation (Figure S7). As the parent $Mg(BH_4)_2$ does not have a peak at such a spectral range, the peak may arise from the HFIP-H related compounds. While it is not clear which components detrimentally affect the electrochemical properties, relatively less amount of (plausibly inorganic) impurities is likely a reason for the advantageous properties of $Mg[B(HFIP)_4]_2$ salt synthesized by transmetalation. Note that, the determination of the exact structure of the impurities is out of our scope.

As described above, the issues associated with the production of $Mg[B(HFIP)_4]_2$ have been resolved. Highly active $Mg[B(HFIP)_4]_2$ can be obtained easily upon combining the corresponding Mg and B compounds. The use of $Mg[B(HFIP)_4]_2$-based electrolytes in MRBs, however, involves other problems.

Reductive decomposition of ether by magnesium: The Coulombic efficiency of the magnesium deposition/dissolution estimated from the CV profiles (Figure S8) and galvanostatic cycling test (*vide infra*) of $Mg[B(HFIP)_4]_2$-based electrolytes reach over 95–98%. The atomic abundance of magnesium in the deposits estimated by EDX analysis was less than 95%. These results are responsible for a side-reaction of magnesium metal with the electrolytes during the deposition/dissolution processes. On the other hand, the open circuit potential of the [Pt || Mg] cell gradually decreased with increasing the storage time (Figure S9). This observation strongly indicates the inherent instability of the electrolytes against magnesium metal even in the absence of any further bias. To assess the long-term compatibility of the $Mg[B(HFIP)_4]_2$-based electrolytes with magnesium metal, a soak test was carried out. Figure 2 summarizes the surface morphology and atomic abundance of magnesium metal soaked in pure G3 and 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 and $Mg[TFSA]_2$/G3 for 3 days at 30 °C.

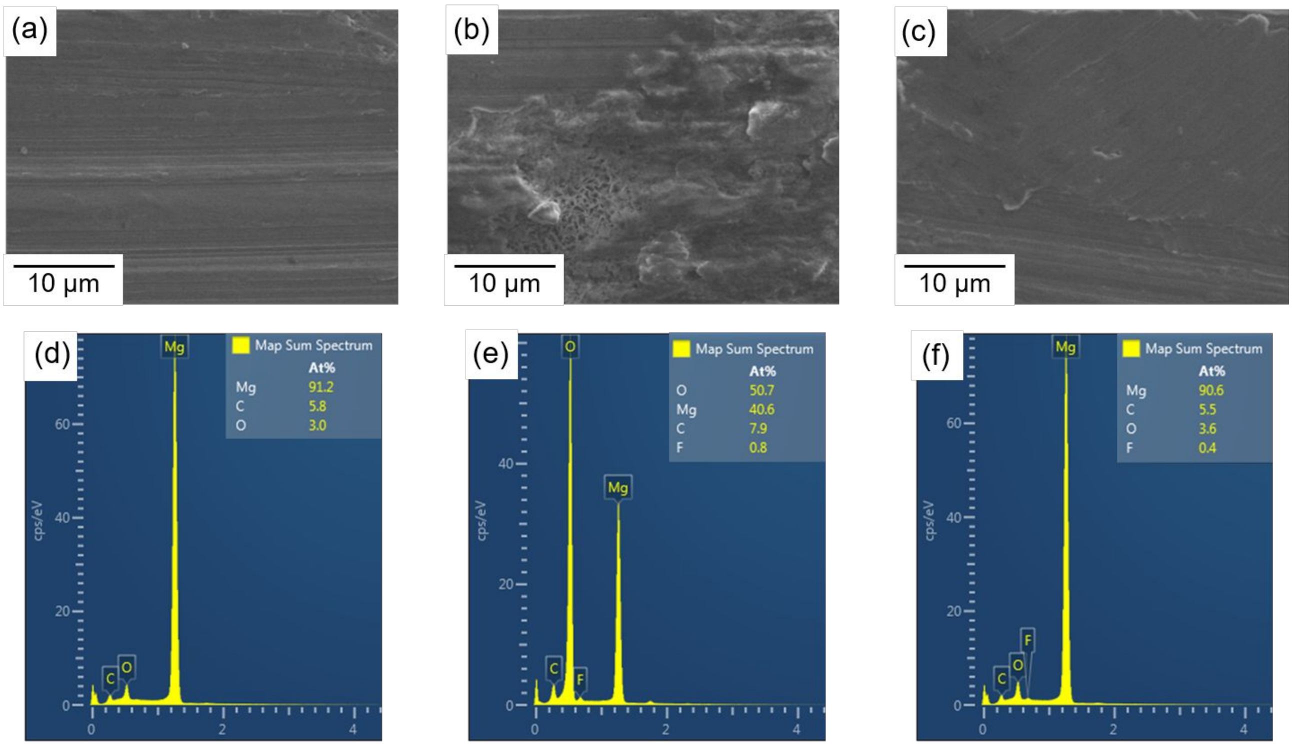


Figure 2. SEM images and corresponding EDX mapping spectra of the magnesium metal soaked in (a) and (d) pure G3 (salt free); (b) and (e) 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3; and (c) and (f) 0.3 mol dm^{-3} Mg[TFSA]$_2$/G3 for 3 days.

As shown in Figure 2, certain traces of the spontaneous side-reaction were observed for the magnesium metal soaked in Mg[B(HFIP)$_4$]$_2$-based electrolytes. The surface of the soaked magnesium metal was unevenly corroded, and sticky reaction products were found on the surface. The internal texture of magnesium can be seen for the soaked magnesium (Figure 2b). The elemental analysis on the observed view and selected areas (Figure S10) indicate that the solvent G3 in Mg[B(HFIP)$_4$]$_2$/G3 is reductively more unstable with magnesium metal than [B(HFIP)$_4$]$^-$ as no or very minor contributions of fluorine were found in the decomposition products. The AES and XPS spectra for the soaked magnesium metal surface also support the absence of the anion contribution to the decomposition products (Figures S11 and S12) as the peaks assignable to MgO, $Mg(OH)_2$, $MgCO_3$, and COC compounds can be seen in the Mg 2p, C 1s, and O 1s XPS spectra for the Mg metal soaked in Mg[B(HFIP)$_4$]$_2$/G3 while no visible peak of MgF_2 was detected in the AES spectra. Although ethers are well-known as reduction tolerant solvents, the reductive decomposition of glymes has been observed in MRB electrolytes; these mechanisms have been discussed intensively so far. Crumlin et al. pointed out the intrinsic instability of diglyme (G2) toward magnesium driven by Mg^{2+} chelation and a subsequent nucleophilic attack by OH^-; both these species, plausibly coming from $Mg(OH)_2$, appeared as impurities on the magnesium metal surface.[26] The Mg^+-mediated solvent decomposition mechanism was also proposed by Persson et al.[27] The passivation of active sites on magnesium metal induced by the accumulation of the decomposition products would, in general, inhibit further decomposition of electrolytes; this is why the degradation of the overall magnesium metal surface is not observed. Such passivation is significantly enhanced with the presence of $[TFSA]^-$.[26,27] The impurities of both electrolytes and electrodes also facilitate $[TFSA]^-$ decomposition,[26,28] thereby concertedly leading to severe passivation. The surface morphology of magnesium metal soaked in pure G3 and $Mg[TFSA]_2$/G3 kept its original characteristics while elements assignable to the decomposition products of the solvent and salt were detected (Figures 2a, 2c, 2d, and 2f). No traces of the undesired reactions were found for the magnesium metal soaked in Grignard reagent 2 mol dm^{-3} C_2H_5MgCl/THF for 3 days at 30 °C (Figure S13). In stark contrast to these previous MRB electrolytes, the decomposition of the electrolyte solvents associated with uneven degradation of magnesium metal likely occurred continuously for the magnesium soaked in $Mg[B(HFIP)_4]_2$/G3. The degradation magnitude of magnesium depends on the concentration of $Mg[B(HFIP)_4]_2$ in the bath electrolytes (Figure S14). These results strongly suggest that not only the surface impurities but the magnesium metal itself reacts with the G3 solvent as a reductant, particularly presented in the $Mg[B(HFIP)_4]_2$/G3 solutions. As no or extremely minor contributions of $[B(HFIP)_4]^-$ were observed, the characteristic coordination state of Mg^{2+} with G3 in these electrolytes induced by the presence of $[B(HFIP)_4]^-$ may cause such phenomena.

The stability of solvents and salts (anions) are strongly related to their coordination state in the electrolyte solutions as the electric field produced by cations induces polarization and consequently affects the oxidative/reductive stabilities of surrounding solvents and anions. The electric field effect is particularly enhanced in cases of small divalent Mg^{2+} because of its high surface charge density.

The change in the LUMO energy levels of $[TFSA]^-$ upon chelate coordination to Mg^{2+} has been investigated intensively, and the reductive stability of $[TFSA]^-$ estimated from DFT calculations agrees well with the experimental stability.[27,29–31] Although the coordination state of $[B(HFIP)_4]^-$ in $Mg[B(HFIP)_4]_2$/G3 is unclear due to the lack of detailed studies on the vibrational characteristics of $[B(HFIP)_4]^-$, a comparison of vibrational spectra between $Mg[B(HFIP)_4]_2$/G3 and $Mg[TFSA]_2$/G3 would provide significant insights for the coordination state of G3 in these electrolyte solutions. Figure 3 displays the Raman spectra for 0.3 and 0.6 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 and $Mg[TFSA]_2$/G3 measured at room temperature. The specific spectral range sensitive to the coordination state of G3 was adopted here. The band observed at 870 cm^{-1} is assignable to complex cation formation.[29,32,33] In the spectra, an intense band assignable to complex formation was observed for $Mg[B(HFIP)_4]_2$ even at the same concentration. This strongly implies that Mg^{2+} in $Mg[B(HFIP)_4]_2$/G3 is preferentially coordinated by G3 and thus, the salt is well dissociated in G3, while both G3 and TFSA anion coordinates to Mg^{2+} in the $Mg[TFSA]_2$/G3 counterpart. The ionic conductivity of these solutions also supports the high dissociativity of $Mg[B(HFIP)_4]_2$ in G3 compared to $Mg[TFSA]_2$: 3.3 and 8.3 mS cm^{-1} for 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 and 1.8 and 3.0 mS cm^{-1} for 0.3 mol dm^{-3} $Mg[TFSA]_2$/G3 at 25 and 80 °C, respectively.

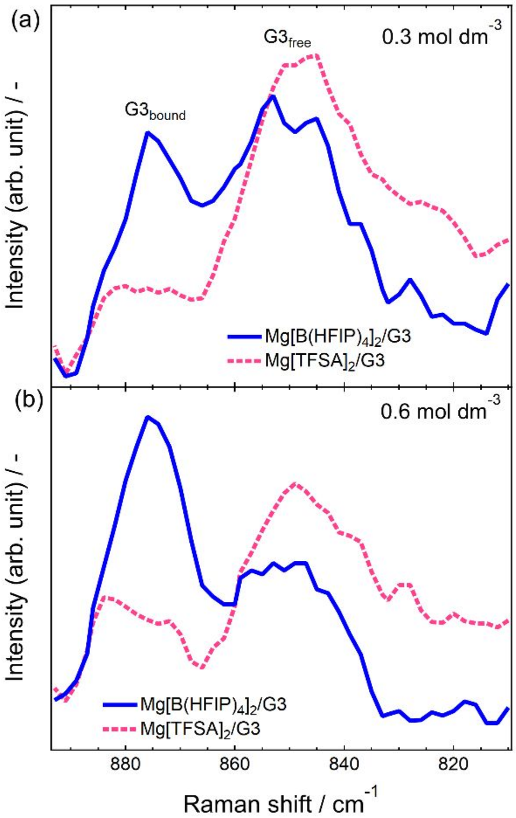


Figure 3. Raman spectra of (a) 0.3 and (b) 0.6 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 and $Mg[TFSA]_2$/G3 measured at room temperature, respectively. The spectral range sensitive to the G3 coordination was adopted.

The LUMO energy level of the isolated (free) $[B(HFIP)_4]$ estimated by DFT calculation using a polarizable continuum model of G3 (ε = 7.62) was −0.78 eV vs. the vacuum level, which corresponds to −1.27 V vs. Mg^{2+}/Mg. Even in contact with Mg^{2+} (CIP, $Mg[B(HFIP)_4]_2$), the level was −0.7 V vs. Mg^{2+}/Mg, indicating $[B(HFIP)_4]^-$ to be stable against magnesium metal irrespective of the coordination states. On the other hand, from the first-principles, the reduction potentials of free uncoordinated glymes were calculated to be below −2.0 V *vs*. Mg^{2+}/Mg; these shifted drastically to positive values *vs*. Mg^{2+}/Mg upon Mg^{2+}-coordination, indicating that coordinated glymes could be reduced by magnesium.[27] The calculation results agreed well with our observations in which solvent G3, instead of $[B(HFIP)_4]^-$, was reductively decomposed by magnesium metal.

Stable solid-electrolyte interface (SEI) formation is one possible approach to achieve the long-term stability of metal anodes in electrolyte solutions. As [TFSA]-based passivation film would suppress the further interfacial reaction (vide supra), magnesium metal was treated with $Mg[TFSA]_2$/G3 electrolyte beforehand then soaked in $Mg[B(HFIP)_4]_2$/G3. The stability of magnesium is improved significantly by this pretreatment. The surface of the pretreated magnesium soaked in $Mg[B(HFIP)_4]_2$ seems smooth (Figure 4a). Subsequent EDX analysis revealed that the pretreatment effectively suppressed the decomposition of G3 (Figure 4b). The XPS spectra indicate the stable protective film formation on the pretreated magnesium. As shown in Figures 4c–4e, the surface chemical composition of the pretreated magnesium soaked in $Mg[B(HFIP)_4]_2$/G3 seems very similar to that of the magnesium soaked in $Mg[TFSA]_2$/G3. Mg 2p spectra showed almost identical spectral features for these two samples as the broad peak located at 50.9 eV diminished with increasing etching time, remaining even after 40 min of argon etching. To further elucidate the surface chemical compositions, the Mg 2p spectra were analyzed and deconvoluted by Gaussian-Lorentzian mixed functions. The deconvolution results were summarized in Figure S15. The deconvolution and subsequent fitting of the Mg 2p spectra of the pristine and pretreated magnesium revealed the presence of two and three components, respectively. The peak observed at 49.7 eV is attributable to metallic magnesium. The components assignable to the native SEI, MgO, $Mg(OH)_2$, and $MgCO_3$ are assumed to be located at higher binding energy of 50.9 eV. Another peak, attributable to MgF_2, was discernible at 51.9 eV for the pretreated magnesium. That peak was also observed for the pretreated magnesium even after soaking into Mg[B(HFIP)$_4$]$_2$/G3. Such peak intensity decreased with Ar sputtering, suggesting the thin protective film formed on the pretreated magnesium. These results combined with the EDX analysis suggests the presence of MgF_2-based protective film on the pretreated magnesium surface, suppressing the undesired decomposition of G3.

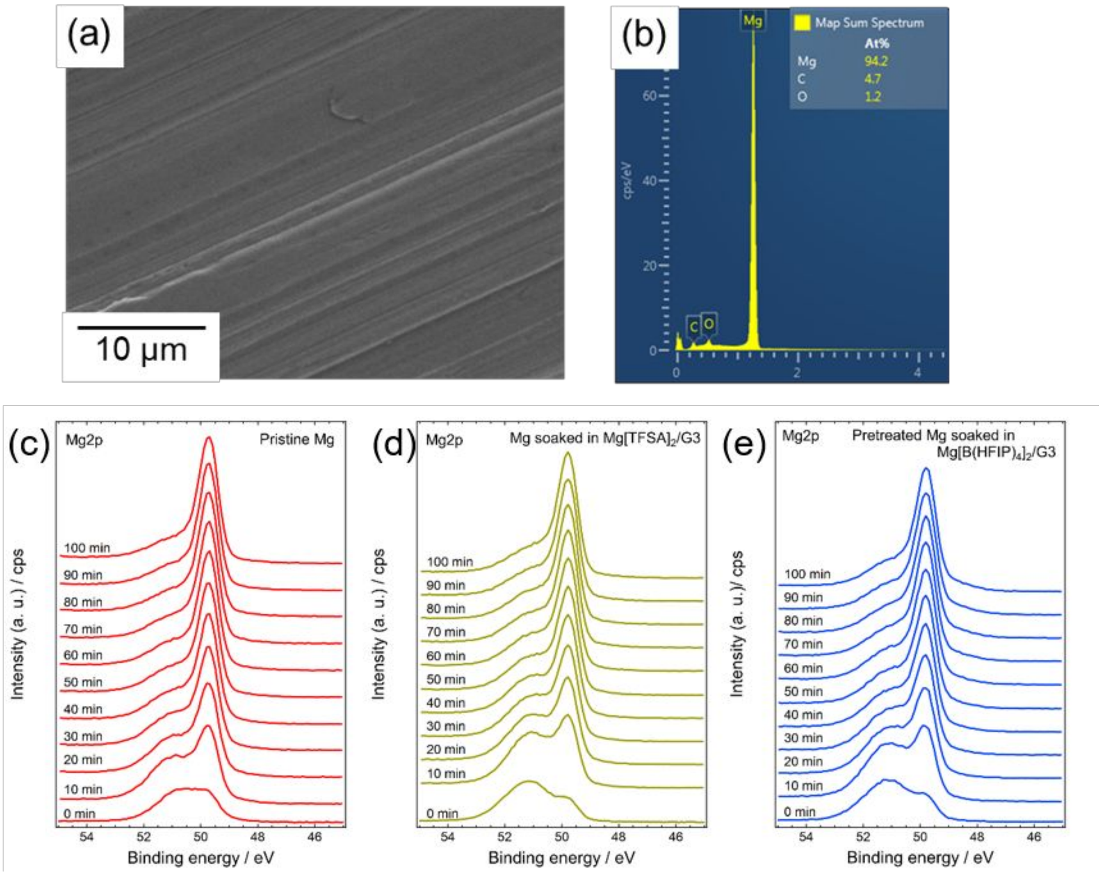


Figure 4. (a) and (b) SEM image and corresponding EDX mapping of the pretreated magnesium soaked in 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$ for 3 days. Depth profiles of (c)–(e) Mg 2p XPS spectra for the non-treated and pretreated Mg metal soaked in either 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3 or 0.3 mol dm^{-3} Mg[TFSA]$_2$/G3 for 3 days, respectively. Time in the XPS spectra indicates the Ar ion beam sputtering times.

It should be noted that the stable magnesium | electrolyte interface was observed for the $Mg(CB_{11}H_{12})_2$/G4 system.[34] Although the detailed coordination state of Mg^{2+} in $Mg(CB_{11}H_{12})_2$/G4 is not clear, the crystallographic data of a series of $Mg(CB_{11}H_{12})_2$/glyme complexes[8] and the extremely low coordinating ability of $CB_{11}H_{12}^{-}$ suggest that Mg^{2+} is preferentially coordinated by G4 in the electrolyte solutions. The soak tests however indicated the opposite observation; stable interface formation was confirmed for the $Mg(CB_{11}H_{12})_2$/G4 while severe magnesium corrosion due to decomposition of glymes was observed for $Mg[B(HFIP)_4]_2$/G3. Moreover, the continuous severe corrosion (not passivation) of magnesium in $Mg[TFSA]_2$/G4 has been reported recently.[34] Therefore, it can be concluded that understanding in the interfacial stability is still controversial and further systematic studies are needed.

The effect of the protective film formation on the electrochemical activity was assessed by cyclic voltammetry of pretreated and non-treated magnesium and subsequent SEM observation. In the initial cycle, the pretreated magnesium showed no current attributable to magnesium dissolution in the anodic scan while the current attributable to magnesium deposition was observed in the reverse cathodic scan despite the required large polarization (Figure S16). After, the magnesium electrode that was plausibly activated and both dissolution/deposition currents could be observed in the subsequent cycles. The non-treated magnesium electrode allowed magnesium dissolution even from the initial cycle. Although the currents observed for the non-treated electrode were larger than those observed for the pretreated magnesium electrode for the initial cycles, the currents were comparable after ten cycles, suggesting that [TFSA]-based protective film on magnesium impedes electrochemical reaction, but gradually relaxing such effects and consequently losing its efficacy. The pretreatment also affects the distribution of active sites. The SEM images of the pretreated and non-treated magnesium metal after 10th CV cycling were shown in Figure S16. The full surface was utilized in magnesium dissolution/deposition cycling for the pretreated magnesium while the limited area was worked for the non-treated magnesium. The full-face passivation of bare magnesium mediated by $[TFSA]^-$ may result in the resetting of active sites on the surface into fully hardly active. This is likely why the full-face passivation was utilized for the pretreated magnesium. Few reports deal with the artificial SEI for MRB applications. Polyacrylonitrile-based artificial film enables the use of carbonate-based electrolytes.[35] The film effectively suppresses the side-reaction between magnesium metal and carbonate while large polarization value of *ca*. 1.0 V was required for magnesium deposition/dissolution even at a very low current density of 0.01 mA cm^{-2}. Recently, Cui et al. successfully constructed a stable SEI layer on the magnesium metal surface via the partial decomposition of certain Li salt, achieving remarkably stable magnesium deposition/dissolution cycling.[36] This approach, however, resulted in hybrid Mg^{2+}/Li^+ electrolytes, which can affect the electrochemistry of cathodes. Alternative Mg-ion conducting SEI and/or surface engineering[37] is needed to materialize MRB based on fully Mg^{2+}/Mg chemistry.

Non-dendritic short circuit: Dendrite growth and subsequent associated issues, such as capacity fade, short-circuits, and fire hazards hinder the commercialization of any practical rechargeable lithium and zinc metal batteries. Magnesium metal, in contrast, does not form dendrite under conventional electrochemical deposition/dissolution conditions. Under mass transfer limitation, magnesium grows dendrite.[38] In this study, the granular deposits can be obtained from the $Mg[B(HFIP)_4]_2$/G3 electrolytes irrespective of the present galvanostatic or potentiostatic conditions in the beaker cells. Short-circuits of the cells, however, occurred in the cells of the two-electrode assembly. The asymmetric two-electrode cells employing Cu working and AZ31 alloy counter electrodes with different separators, where an amount of 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 was injected, were assembled in the Ar filled glove box. The galvanostatic cycling behavior depended on the separator used. The cells with GA200 achieved stable magnesium deposition/dissolution cycling over 100 cycles while short-circuiting occurred for cells with GF/A and 3DOM separators at 33 and 7 cycles, respectively (Figure 6). The deposition/dissolution overpotential and Coulombic efficiency of the cells with GA200 on Cu were ca. 180/110 mV and ca. 92%, respectively, for the initial cycles, reaching up to ca. 50/50 mV and 98%, respectively, for subsequent cycles. As GA200 and GF/A separators are both made from glass fiber thus have similar microstructures, the compatibility of electrolytes with glass fiber is not a likely reason for the short-circuiting observed for the cells with GF/A; the thickness of the separator would determine the cycle number.

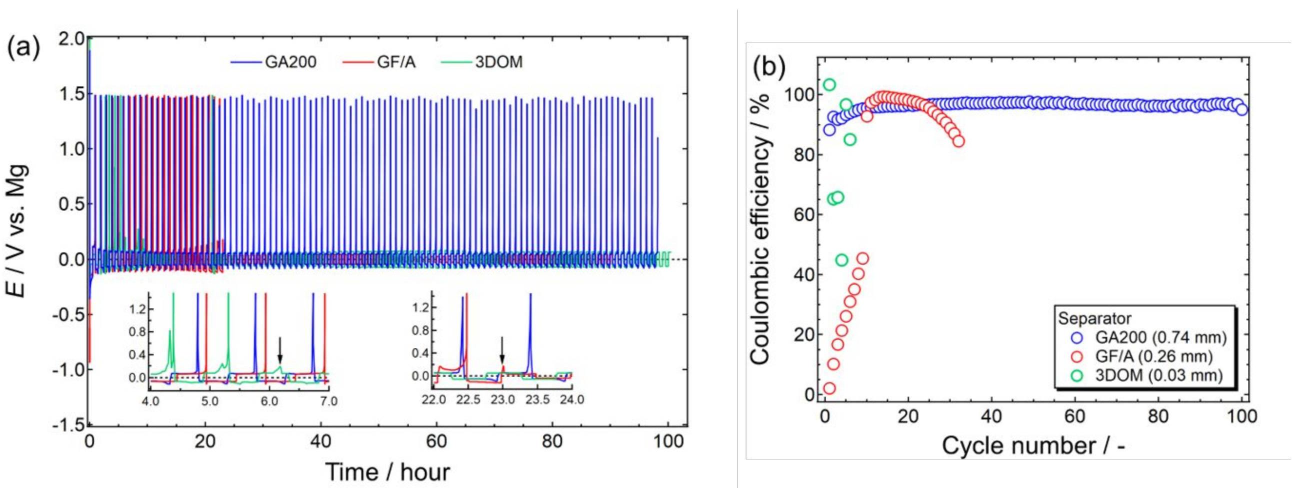


Figure 5. (a) Galvanostatic deposition/dissolution cycling and (b) Coulombic efficiency of [Mg | 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3 | Cu] cells using three different separators, GA200, GF/A, and 3DOM. The cycling tests were carried out at a current density of 0.5 mA cm^{-2} and 30 °C. (a) Allows in the insets point out when short-circuiting would take place.

To clarify the physical cause of short-circuits, the cycled cells were disassembled and the separators were observed by SEM. The digital images of the cycled separators indicate magnesium deposition on the Cu | separator interface and penetration of the deposits into the separators (Figure 6). SEM observation on the cross-sectional view and subsequent EDX analyses further corroborate the deposits remaining in the microstructure of the glass fiber separators (Figure S17). The penetration and isolation of magnesium particles were seen in the cross-sectional view of the 3DOM separator retrieved from the cell cycled only once (Figure S18). 3DOM separator has the uniform inverse opal-type microstructures with pores of approximately 300 nm diameter.[20] Owing to its well-controlled microstructure and resulting homogeneous mass transfer in the separator, dendrite growth as well as undesired penetration and resulting short-circuiting can be effectively suppressed during lithium deposition/dissolution cycling for over 400 times even at a current density of 1.0 mA cm^{-2} in the certain case.[39] Even in the conventional lithium electrolytes, 3DOM separator allows improved cycling compared to the conventional polypropylene-based separators.[20] Despite such outperforming feature of the 3DOM separator, short-circuits took place in several magnesium deposition/dissolution cycling even though relatively low current density applied (0.5 mA cm^{-2}). For the glass fiber separators GF/A and GA200, they are capable to capture certain amount of the deposited magnesium particles that isolated from the ones deposited on the electrodes. Such isolation causes less cycling efficiency. The uneven utilization of magnesium (that coming from the surface nature of magnesium metal) and remarkably poor affinity of magnesium deposits on metal substrates are likely reason for the observation of the penetration and isolation of deposits into the separator. The remained deposits on the other hand can cause short-circuits, as they can tie each other from the anode to cathode sides upon repeated deposition/dissolution cycles. Continuous deposition in the separator can also lead to short-circuits. Therefore, short-circuiting for the cells is induced by the physical connection among the isolated and continuous deposits upon cycling; this is why cells with thicker separators can be cycled more times than those with thinner separators. Indeed, the cells with GA200 separator allow further long-term magnesium deposition/dissolution cycling over 250 times (Figure S19). The thickness of GA200 seems however inappropriate for practical MRB materialization. To achieve stable, long-term magnesium deposition/dissolution cycling, separators having further small pore size and/or magnesiophobic nature are needed.

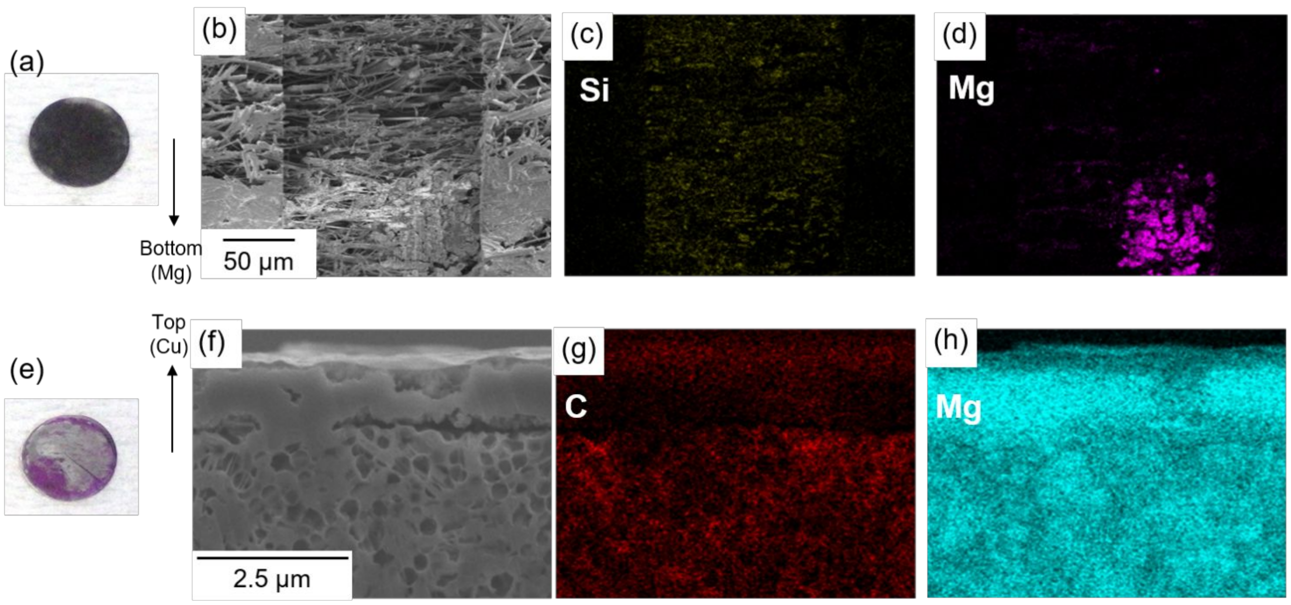


Figure 6. Digital and SEM-EDX images of the cycled (a)–(d) GA200 and (e)–(h) 3DOM separators. (a), (e): the top surface is the side in contact with Cu.

It is paramount to reveal the primary cause of the isolation of deposits in separators to develop rational pretreatment techniques and/or materials for practical MRB applications. As described in the former section, the active sites seem to be distributed unevenly on the magnesium metal surface. The cycled magnesium anode also showed a characteristic feature where the surface was unevenly utilized (Figure S20). Such uneven utilization became prominent when elevating the temperature. The magnesium anode cycled at 60 °C lost its original shape due to extremely localized utilization and many holes were found on the overall surface (Figure S20). The recent systematic studies of the temperature effects on the lithium dendrite pointed out the double-edged effect of temperature – the uniform thermal field leads to dendrite-free smooth surface while the localized hot spot rather induces dendrite growth.[40] Although the intrinsic chemical nature of magnesium is different considerably from that of lithium, a similar mechanism is considered to be applied in the magnesium deposition/dissolution behavior as the active sites, namely hot spots, are unevenly distributed due to uneven surface passivation by native SEI. Highly active sites are supposed to be utilized preferentially at higher temperature owing to the enhanced mass transfer. Such localized uneven utilization of magnesium anodes would concertedly facilitate a short-circuit (Figure S21). For magnesium metals, the atomic packing density governs the corrosion behavior.[41] This indicates that the corrosion behavior is surface-orientation dependent and the surface orientation determines the distribution of active sites on the magnesium metal surface. Therefore, manipulation of the active site distribution on magnesium anode surface based on the metallography is crucial in preventing undesired short-circuits and to realize stable electrochemical Mg^{2+}/Mg cycling.

Conclusion

The issues of fluorinated alkoxyborate-based electrolytes for practical MRB applications and possible strategies to resolve them were described. The previously reported general synthetic route of $Mg[B(HFIP)_4]_2$ using $Mg(BH_4)_2$ causes batch-dependent electrochemical activity as well as an insufficient synthetic yield of the salt. The alternative procedure based on the transmetalation between the corresponding magnesium fluorinated alkoxide and borate led the highly active $Mg[B(HFIP)_4]_2$ with very high reproducibility in the synthetic yields and electrochemical performance. Despite the outstanding electrochemical activities of $Mg[B(HFIP)_4]_2$/glyme, the electrolytes were found to be unstable against magnesium metal. Magnesium metal soaked in $Mg[B(HFIP)_4]_2$/glyme was unevenly corroded and the decomposition products, composed mainly of Mg, C, and O, were found on the surface, suggesting that solvent glymes not $[B(HFIP)_4]^-$ were reduced by magnesium. The remarkably high dissociativity of $Mg[B(HFIP)_4]_2$ in glyme solutions and the resulting enhanced induction interaction of bare Mg^{2+} with the surrounding glymes would make the reduction potential of coordinated glymes positive vs. Mg^{2+}/Mg. Surface passivation by [TFSA]-based electrolytes worked well to suppress the decomposition of $Mg[B(HFIP)_4]_2$/glyme electrolytes, while simultaneously causing a large overpotential for electrochemical cycling. Further investigations aimed at controlling the coordination states of solvents and anions, as well as the interface engineering, are needed. The short-circuit of the cells upon repeated deposition/dissolution cycling is rather problematic for MRB materialization. Uneven distribution of active sites on the base magnesium surface and repeated electrochemical cycling concertedly cause highly localized utilization. Such uneven utilization provokes the isolation of deposited magnesium particles in the microstructure of separators due to the loss of electrical connection to the current collector; consequently, a short-circuit is induced by the percolation of those isolated particles in addition to continuous localized deposition. The findings here disclose the issues of fluorinated alkoxyborate-based electrolyte solution that should be resolved for practical MRB applications and propose the importance of systematic strategies in manipulating the electrolytes, interfaces, and base magnesium metal based on each appropriate approach.

Supporting Information.

^{1}H NMR spectra of the synthesized $Mg[B(HFIP)_4]_2$ and possible intermediates; CVs of Mg electrodes recorded in 0.3 mol dm^{-3} $Mg[Al(HFIP)_4]_2$/G2; CVs of 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 containing impurities; CVs of 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G*n* (*n* = 1–3) and $Mg[B(HFIP)_4]_2$/THF; initial and 10th CVs of 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; SEM images of deposited magnesium metal from 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; Raman spectra of the synthesized $Mg[B(HFIP)_4]_2$; Charge passed during cycling in 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; Time profiles of open circuit potential of the [Mg | 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3 | Pt] cell; SEM images and EDX mapping of magnesium metal soaked in 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; SEM images and AES spectra of magnesium metal soaked in 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; Depth profiles of pristine magnesium metal and that soaked in 0.3 mol dm^{-3} $Mg[B(HFIP)_4]_2$/G3; SEM image and EDX mapping of magnesium metal soaked in 2 mol dm^{-3} C_2H_5MgCl/THF; SEM images and EDX mapping of magnesium metal soaked in 0.1, 0.3, and 0.5 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3; Mg 2p spectral deconvolution results of pristine magnesium and those soaked in 0.3 mol dm^{-3} Mg[TFSA]$_2$/G3 and Mg[B(HFIP)$_4$]$_2$; CVs of pretreated magnesium metal recorded in 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3; SEM images and EDX mapping of the cross-sectional view of GA200 separator after cycling; SEM images and 3D EDX mapping of 3DOM separator after cycling; Galvanostatic deposition/dissolution cycling of [Mg | 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G2 | Cu] cell; SEM images of magnesium negative electrode after cycling; Galvanostatic deposition/dissolution cycling of [Mg | 0.3 mol dm^{-3} Mg[B(HFIP)$_4$]$_2$/G3 | Cu] cells at different temperature.

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Graphical Abstract